How Fast Does the Lightning Network Patch?

Release Lines from Gossip and Exposure after Security Advisories, 2023–2026

Saurabh Kumar
Bitcoin Data Labs
sorukumar@bitcoindatalabs.org

***Abstract*—Every Lightning implementation disclosed security vulnerabilities in 2026, and every advisory told operators to upgrade. Whether they do has never been measured, because Lightning gossip carries no software version. We show that the release line of a public node can be read from the feature bits it announces: matched against the source of every release of LND, Core Lightning, Eclair and LDK, the inferred range contains the true line on 99.99% of 15,549 node-days of nodes that state their version, and the most likely line is exact on 98.2%. Applying it to 814 daily snapshots of the public graph from 2023 to 2026, we find that half of the LND nodes on an older line upgrade within 90 days of a release, in waves consistent with node platforms pushing updates, while small nodes and nodes that skipped earlier releases lag far behind. One operator ran up to 1,781 public LND nodes, 28% of all live LND nodes, with about one bitcoin between them. For 30 LND and CLN advisories, fixes shipped a median 256 days before disclosure. LND's disclosures did not change how fast operators left vulnerable releases: two months after Lightning Labs' August 2026 batch, 71% of the nodes exposed at disclosure still ran an affected release. CLN's disclosures of July 2026 raised the monthly rate at which nodes left vulnerable releases two- to fourfold. Exposure measured by capacity is small; by node count, it depends on how one counts a single operator's fleet.**

***Index Terms*—Lightning Network, Bitcoin, patch deployment, vulnerability exposure, software versions, measurement, gossip**

I. Introduction

Between July and October 2026, every major Lightning implementation disclosed security vulnerabilities. Lightning Labs published ten LND advisories on a single day [1] and eight more a month later [2]; Core Lightning (CLN) asked operators who could not upgrade to restart their nodes offline [3]; Eclair and LDK shipped security releases. Each advisory ends with the same advice: upgrade. Whether operators do, and how fast, decides how long a disclosed vulnerability stays exploitable. For web servers, browsers and Bitcoin and Ethereum full nodes, this question has been measured for two decades [4]–[8]. For the Lightning Network it has not, because Lightning gossip carries no software version: earlier studies could tell which implementation a node runs [9]–[11], not which release.

We show that the release can be read from gossip after all, and measure what operators do with it. Each release announces a characteristic set of protocol feature bits. From the source of every release of LND, CLN, Eclair and LDK since 2019 we derive which sets each release can produce, and match them against 814 daily snapshots of the public graph taken by our own node from May 2023 to October 2026. We make four contributions.

- **Release-line inference from gossip.** We infer the release line (the finest unit gossip separates; patch releases are invisible) of every public node with a known implementation. On 124 CLN nodes whose alias states their version, the inferred range contains the true line on 99.99% of 15,549 node-days and the most likely line is exact on 98.2% (§IV).
- **Upgrade dynamics.** New LND lines reach half of live nodes in 72–88 days. Half of the LND nodes on an older line upgrade within 90 days of a release, but the smallest quarter of nodes takes 142, and a node already more than one line behind has about five times lower odds of moving in any week. Upgrades arrive in waves consistent with node platforms pushing updates (§V).
- **A managed fleet.** One operator ran up to 1,781 public LND nodes, 28% of all live LND nodes, with about one BTC of capacity between them, and upgraded them in step. Counted as independent nodes, they would distort every node-based statistic for 2025–2026, including adoption and exposure.
- **Exposure after advisories.** For the 30 LND and CLN advisories whose fix boundary gossip can see, we measure how many live nodes and how much capacity remained exposed. Fixes typically shipped months before disclosure (median 256 days). LND disclosures did not change how fast operators left vulnerable releases: after Lightning Labs' August 2026 batch, 71% of the exposed nodes were still live on an affected release two months later. CLN operators did respond: the share leaving vulnerable lines in the 30 days after CLN's July 2026 disclosures was two to four times that of the 30 days before (§VI).

All data behind the figures and tables, and the code, are released. We report aggregates only: no list, ranking or label of individual nodes by release or exposure.

II. Background and Related Work

A. Gossip and feature bits

Public Lightning nodes announce themselves and their channels through gossip [12]. A node's `node_announcement` carries its alias, colour, network

addresses and a feature vector: a bit field in which each pair of bits marks a protocol feature as required (even bit) or optional (odd bit) [13]. Implementations set their feature vector from compiled-in defaults, which change when a release adds or retires a feature, plus whatever options the operator switches on. Gossip carries no version string. A node re-signs its announcement when it restarts or changes its settings, so a node that upgrades and restarts re-announces with the feature vector of the new release.

B. Inferring implementations

Zabka et al. classify public nodes into LND, c-lightning and Eclair from default channel parameters, colours and aliases [9], [14]. Their study contains the only prior observation of an LND upgrade we found: after LND changed its default CLTV delta from 144 to 40 in March 2019, the new value overtook the old one around mid-December, about nine months later. They name the planning of critical software updates as a use of implementation inference but do not measure it. Espinasa-Vilarrasa et al. infer the implementation from gossip traffic or from announced feature bits with decision trees [10]. Kappos et al. note that their gossip snapshots carried no version information, and fell back on owner-reported fields for 370 nodes [15]. Our previous paper used feature-bit rules to measure client concentration and the 2026 Core Lightning shutdown [11]. All of these stop at the implementation; none infers which release a node runs.

Longitudinal studies of the Lightning graph cover topology, centralisation and robustness [16]–[20], infrastructure [21], privacy [22], channel lifecycles [23], gossip convergence [24] and, most recently, stability and node lifetimes [25]–[27]. None of them studies software versions. When Lightning Labs disclosed CVE-2019-12999, they were "surprised to find many nodes running very old versions" but published no numbers [28].

C. Versions in other blockchain networks

Bitcoin and Ethereum peers report their software version in the handshake, so crawlers can count versions directly. Kim et al. found 68% of Geth nodes two or more releases behind and tracked client age against release dates [8]. Neudecker reports that Bitcoin nodes adopt a release mostly in its first two months, and documents an alternative client that went from zero to more than 700 peers in a day [29]; Park et al. and Eisenbarth et al. survey Bitcoin node versions and their known vulnerabilities [30], [31]. SRLabs estimated that about 40% of Parity nodes were still vulnerable a month after an advisory [32]. Zcash sidesteps slow upgrades by expiring old releases at network upgrades [33]. The Lightning Network offers neither a version string to count nor a forced expiry.

D. Patch deployment

The window of vulnerability, from disclosure to patch, was framed by Arbaugh et al. [4]. Rescorla probed servers affected by the OpenSSL vulnerabilities of 2002 and found more than two thirds still vulnerable two weeks after disclosure [5]. Frei et al. showed that automatic updates dominate how fast a client population patches [6], Durumeric et al. followed Heartbleed across the Internet [7], and Nappa et al. modelled patch deployment with survival analysis [34]; Li and Paxson measured the life cycle of security fixes from the code side [35]. We bring this lens to a network where versions must first be inferred, and where a release is often fixed months or years before the vulnerability is disclosed.

III. Data and Vantage Point

a) Gossip snapshots: We take a daily dump of the public graph (`describegraph`) from our own LND node and keep, per day, every node announcement (feature bits, alias, colour, addresses, last-update time) and every channel direction (the policy a node sets for forwarding towards its peer, with capacity, the *disabled* flag and last-update time). The snapshots run from May 2023 to the data cutoff of 13 October 2026. A per-day quality check discards dumps that are partial (fewer than half the channels of their period), stale (fewer than half the channels updated within two weeks) or error responses written while the node's wallet was locked; 814 days remain. Daily coverage is dense from November 2023, with gaps in May–June 2024 (partial dumps), August 2024 and January–May 2025 (wallet locked after restarts). Upgrade times observed across a gap are known only to lie inside it; we treat them as interval-censored (§V).

b) Vantage: All snapshots are one node's view. LND prunes channels that have not been updated for about two weeks, and our view widened in July 2024 (from about 8.4k to 11–13k nodes with channels), after the node's own peer set grew. We compare levels within a period and report shares, which are less sensitive to the view; in particular, the few live nodes on releases from before 2021 appear only from July 2024. Only announced (public) nodes are visible: mobile wallets and services behind unannounced channels, where much of LDK and Eclair runs, are not.

c) Who is still there: Nodes that stop running do not leave the graph at once: their peers mark the channels towards them disabled, and the channels stay announced until they are closed or pruned. An offline node cannot announce anything, so liveness must be read from its peers' side [11]. We call a node *live* on a day when at least one channel direction towards it, set by a peer, is enabled and was updated in the previous 14 days; its *live capacity* is the capacity of those channels. We also report 7- and 28-day variants. The distinction matters. On the cutoff day, 2,689 LND nodes with channels (outside the managed fleet of §V) announced a release older than 0.19, but only 434 of them were live; counting every announced node would put the share of LND nodes on those releases at 42.3%, against 12.5% of live nodes.

IV. Inferring Release Lines from Gossip

A. Release lines

We built a table of the `node_announcement` feature bits of every release of the four implementations from their source at each release tag: LND 0.7–0.21.4, Core Lightning (CLN)

0.8.2–26.06.9, Eclair 0.3–0.14.3 and LDK 0.0.113–0.2.7. For each release we record the bits it sets by default, the bits an operator can add through options, flags, plugins or known custom extensions, and the default bits an operator can switch off; each entry cites the source file it comes from. Consecutive releases with identical defaults form a *release line*, the finest unit gossip can resolve: LND has 10 lines from 0.8 to 0.21, CLN 13, Eclair 13 and LDK 6. Some lines are single minor releases (LND 0.18, 0.19, 0.20, 0.21); others span several (LND 0.15.1–0.17; LDK 0.0.120–0.1 covers January 2024 to December 2025). Patch releases never change the defaults, so a patch level is never visible.

For a node with feature set B, a line ℓ with defaults D_ℓ, optional bits O_ℓ and removable bits R_ℓ is *consistent* when $D_\ell \setminus R_\ell \subseteq B \subseteq D_\ell \cup O_\ell$. The *strict range* is every consistent line released by the day of observation; it contains the running line under any non-developer configuration. The *likely line* is the consistent line that needs the fewest non-default bits among the lines released by the day the node's current feature set first appeared. Dating the inference to that day matters: a node cannot run a line that did not exist when it began announcing its bits, and without it a new release would relabel every node whose existing bits happen to fit the new line (233 CLN nodes running 24.05 with the onion-message option would have "upgraded" to 24.08 on its release day). A set consistent only with lines not yet released is a pre-release build; a set consistent with no line is unrecognised.

B. Implementation

Release lines are defined per implementation, so the implementation comes first. We extend the feature-bit rules of our previous paper [11]: LND-only bits (2023, 31) and Eclair-only bits (37, or 29 with 61) are decisive; a set with bits 11 and 39 or 43 is CLN when bit 5 is set and Eclair otherwise; 39, 43 or 63 without 11 is LDK. New in this paper, the release table itself is evidence. When a set is consistent with the lines of exactly one implementation, that implementation is assigned, which recovers releases older than the distinguishing bits, such as LND 0.8–0.11 and CLN before 23.05; and when a bit rule names an implementation that has no release producing the set while exactly one other implementation does, the table wins. Weak signals (LND's default colour or alias, default CLTV deltas) label the remaining nodes with low confidence, and only operator-attributed and high-confidence labels receive a release line. Announcements that are not visible on a day (last update 0) carry the node's previous feature set forward.

C. Validation

a) Implementation: 797 nodes name their implementation in their alias ("…[LND]", "…-CLN", "eclair", "ldk-node"); LND's default alias is excluded because the rules use it. On the latest day each was announced with channels, the labels reach precision 100.0% and recall 96.4% for LND (664 nodes) and precision 98.3% and recall 99.1% for CLN (117 nodes) (Table I). The misses are almost all low-confidence labels, which receive no release line.

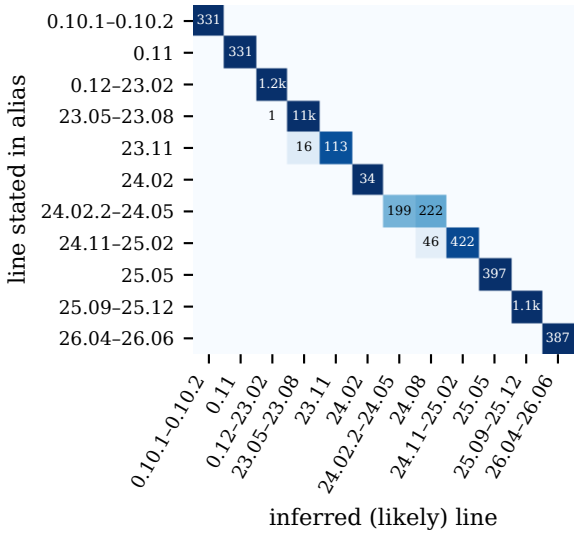


Fig. 1. Release line stated in CLN developer aliases (rows) against the likely line inferred from feature bits (columns), in node-days. Shading is the row share.

b) Release line: CLN built in developer mode puts the version into the default alias (e.g. `YELLOWWATER-v25.09`); the alias follows the binary, so it is correct on every day the node is announced. Over 15,549 announced node-days of 124 such nodes on 11 lines (no operator above 5% of node-days), the strict range contains the stated line on 99.99% of node-days and the likely line is exact on 98.2% (Fig. 1). Of the 136 combinations of node and stated line, 132 are exact on every day. The misses are a 24.05 node with the onion-message option, which makes it identical to 24.08; a 25.02.1 node and a 23.11.1 node whose switched-off options make them look one line older; and a single day of a release candidate. For LND, versions stated in aliases are rare and set by hand; we use them only as a spot check.

c) What gossip cannot resolve: Among live nodes with a release line, the strict range spans more than one line on 11% of LND and 47% of CLN node-days in 2026, because options let an older line imitate a newer one; the likely line is unique on all of them. Unrecognised sets are 1.0% (LND) and 0.1% (CLN) of node-days, and pre-release builds about 0.1% of LND node-days. Three limits remain. Patch releases are invisible. A node that upgrades without restarting, or whose new announcement does not reach us, keeps its old bits. Developer overrides and forks can imitate any set. The first two bias exposure upward, the third in either direction; none can be measured from gossip alone, which is why we report the strict and likely ranges side by side.

V. How Lightning Nodes Upgrade

A. One operator, many nodes

Before measuring upgrades, one population has to be set apart. From 20 June 2025, hundreds of LND nodes appear with aliases of the form `LNT.<name>`, one or two channels

TABLE I
VALIDATION AGAINST SELF-IDENTIFYING ALIASES. TOP: CLIENT LABELS OF 797 NODES WHOSE ALIAS NAMES THEIR IMPLEMENTATION. BOTTOM: RELEASE LINES OF CLN NODES WHOSE ALIAS STATES THE RUNNING VERSION, GRADED ON EVERY ANNOUNCED NODE-DAY.

Client	Nodes	Precision (%)	Recall (%)
LND	664	100.0	96.4
CLN	117	98.3	99.1
Eclair	9	58.3	77.8
LDK	7	100.0	100.0
CLN nodes / lines with a stated version			124 / 11
Node-days graded			15,549
Strict range contains the stated line (%)			99.99
Likely line equals the stated line (%)			98.2

each, most of them to a handful of hubs with the same prefix, and identical Lightning Terminal feature sets. At its peak on 7 December 2025 the fleet ran 1,781 nodes, 28% of all live LND nodes, with 1.3 BTC of live capacity against 8,300 BTC for live LND as a whole; at the cutoff it still ran 566 nodes (14% of live LND nodes). The fleet upgrades as one: 327 of its nodes moved to LND 0.20 on 9 December 2025. Its upgrades are one operator's decisions, so we exclude its nodes (by alias prefix, or by having at least half their channels to a fleet hub) from every per-node statistic below and describe them here instead. The fleet also illustrates why node counts are a poor measure on the Lightning Network [11]: for half a year, more than a fifth of the live LND nodes we observe belonged to one operator and held less than a thousandth of LND's live capacity.

B. Adoption of new release lines

Fig. 2 shows the release-line mix of live LND and CLN nodes. Each new LND line displaces its predecessor within a few months but leaves a tail: at the cutoff, 12.5% of live LND nodes were still on a line older than 0.19, released in May 2025. New lines reached half of live LND nodes 72–88 days after release (Fig. 3); half of live capacity moved as fast or faster for 0.18 and 0.19 (41 and 69 days) but more slowly for 0.20 (127 days). Counting the fleet as independent nodes would put LND 0.20 at 50% of nodes after 54 days instead of 88. CLN lines took 66–216 days. Every LND line from 0.18 on was announced by a few nodes 53–174 days before its release: developers and early adopters running unreleased builds on mainnet.

C. Time to upgrade

Adoption curves mix upgrades with nodes arriving and leaving. To follow operators, we take every node that is live on an older line on the release day of LND 0.19, 0.20 or 0.21 or of a CLN line from 23.11 on, and measure the time until it is first seen on the new line or a later one. Snapshot gaps leave the upgrade day inside an interval (for 86% of LND upgrades it is at most two days wide); nodes that stop being live before upgrading are censored at their last live day.

TABLE II
WHAT PREDICTS UPGRADING: DISCRETE-TIME LOGIT ON NODE-WEEKS AFTER EACH LND RELEASE (WEEK AND RELEASE FIXED EFFECTS; STANDARD ERRORS CLUSTERED BY NODE). ODDS RATIOS WITH 95% CI.

Covariate	Odds ratio	95% CI
Capacity ($\times$10)	1.11	[1.07, 1.16]
Channels ($\times$10)	1.15	[1.06, 1.24]
Tor-only	1.59	[1.50, 1.69]
LND default colour	1.08	[1.03, 1.14]
Seen before 2024	1.01	[0.96, 1.07]
$>$1 line behind	0.19	[0.18, 0.21]
Nodes / node-weeks		7,218 / 228,254

The share upgraded by day t is the nonparametric maximum-likelihood estimate for interval-censored data [36]. Half of the LND nodes upgraded within 90 days and 76% within a year; for CLN the median is 137 days. Size matters at the bottom: the smallest quarter of LND nodes by live capacity took a median 142 days, the largest quarter 79 (Fig. 4).

Table II fits a discrete-time model of the weekly probability of upgrading for LND. Larger nodes upgrade sooner (odds ratio 1.11 per tenfold capacity, 1.15 per tenfold channel count), and so do Tor-only nodes (1.59). A node more than one line behind when the release ships is far less likely to move (0.19): the tail is made of nodes that skipped releases before. Nodes with LND's default colour are no slower (1.08), and nodes seen before 2024 upgrade at the same rate as newer ones (1.01).

D. Upgrade waves

Upgrades arrive in bursts (Fig. 5). Outside the fleet, on 9 days at least 114 LND nodes, and up to 252, moved to a new line, 23–69 days after its release. These are not new or empty nodes: their median live capacity (0.051 BTC, against 0.053 BTC for other upgraders of the same line) and their median age in our data (471 days) rule out a fleet of fresh nodes. 8 of the waves are more often Tor-only than the network (76–93% of the wave, against 60–73% of all live LND nodes), and about half of each shares one feature set: the pattern of node platforms that ship an update to all their users at once, though gossip cannot name the platform. The remaining wave is different: 114 mostly clearnet nodes (26% Tor-only) moved to LND 0.21 on 8 August 2026, the day after BTCPay Server published its security release 2.4.2, which bundles LND 0.21.1 [37].

VI. EXPOSURE AFTER SECURITY ADVISORIES

A. Advisories gossip can see

We collected every public security advisory for LND, CLN, Eclair and LDK from January 2022 to the cutoff (release notes, Lightning Labs' advisory index, Delving Bitcoin and authors' write-ups), recording the affected versions, the first fixed release and the disclosure date. An advisory is visible in gossip when its fix boundary coincides with the start of a release line (LINE), or falls inside a line but leaves whole lines affected

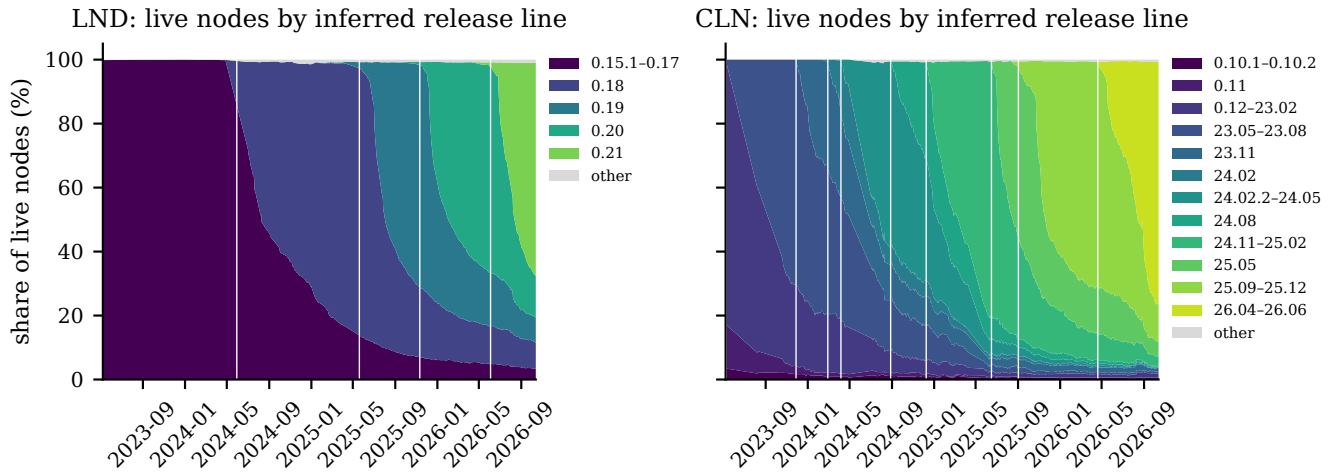


Fig. 2. Live LND and CLN nodes by inferred release line (likely line), excluding the managed fleet. White lines mark releases. Lines under 2% of nodes throughout are grouped as "other".

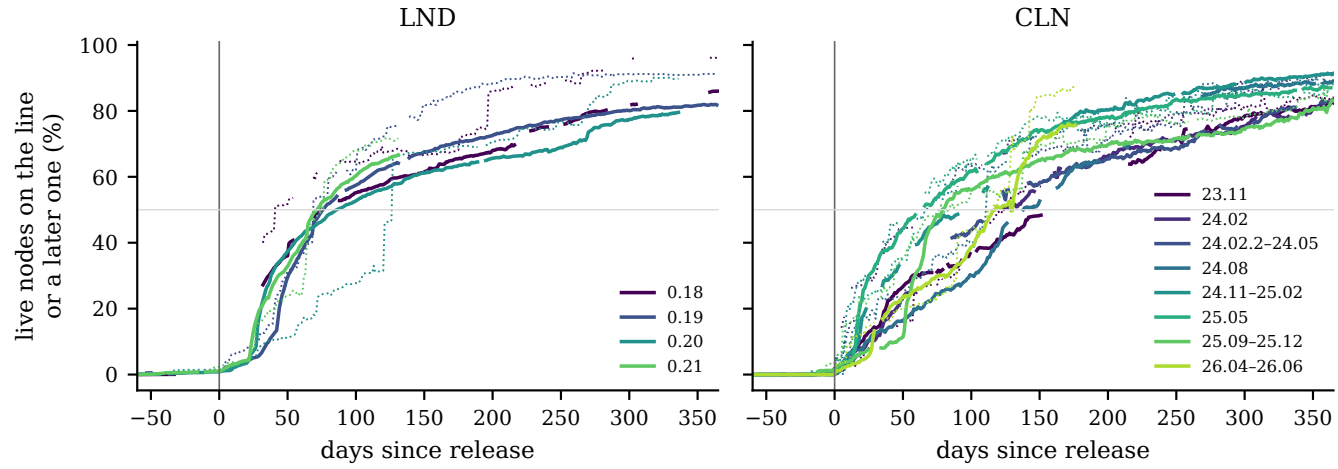


Fig. 3. Share of live nodes (solid) and of their live capacity (dotted) on each release line or a later one, by days since the line's release. Lines are broken across snapshot gaps longer than three days.

or fixed on either side (PARTIAL, which gives bounds). Fixes inside the newest line, such as CLN 26.06.7–26.06.9 and Eclair 0.14.1–0.14.3, are patch releases and invisible. Eclair and LDK have too few live public nodes with a known line (about ten Eclair nodes) for shares to be meaningful, so Table III lists the 23 LND and CLN entries, covering 30 advisories.

Two clocks run for each advisory. Fixes usually ship long before the vulnerability is disclosed: the median gap between the first fixed release and disclosure is 256 days, and the six advisories Lightning Labs published on 11 August 2026 concerned bugs fixed in LND 0.19.0, released 446 days earlier. A node still on an affected line at disclosure has therefore usually skipped a release that was already available for many months. Exposure is the share of live nodes (excluding the managed fleet) on an affected line, with the share of their live capacity alongside.

B. Disclosure barely moves LND operators

On the day Lightning Labs disclosed its batch, 14.9% of live LND nodes, holding 6.4% of live LND capacity, were on a line older than 0.19 (Fig. 6). Of the 550 exposed nodes, 54 had upgraded by the cutoff, 108 were no longer live and 388 (71%) were still live on an affected line. The disclosure did not change the pace: in the 30 days before it, 6.9% of the live nodes then on affected lines moved to a fixed line; in the 30 days after, 6.2%. Morehouse's disclosure of three LND vulnerabilities fixed in the same release, on 4 December 2025, looks the same: 27.7% of live LND nodes exposed, an upgrade rate of 5.8% before and 6.5% after, and 55% of the exposed cohort still exposed four months later. The exposed nodes are small: their share of capacity is about half their share of nodes.

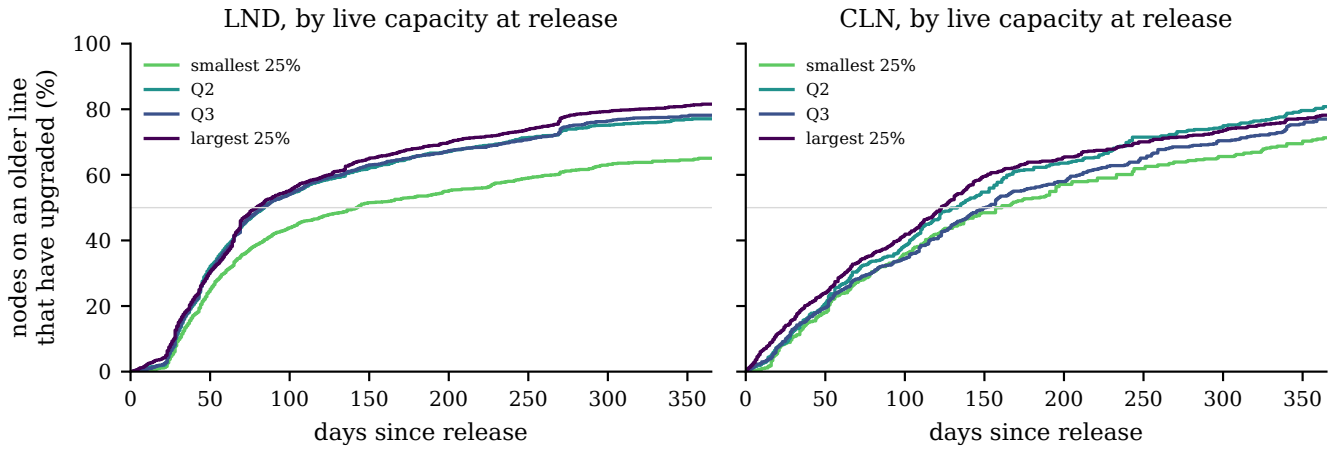


Fig. 4. Share of nodes that were live on an older line at a release and have since been seen on the new line or a later one, by quartile of live capacity at the release (Turnbull estimate; LND 0.19–0.21, CLN 23.11–26.04).

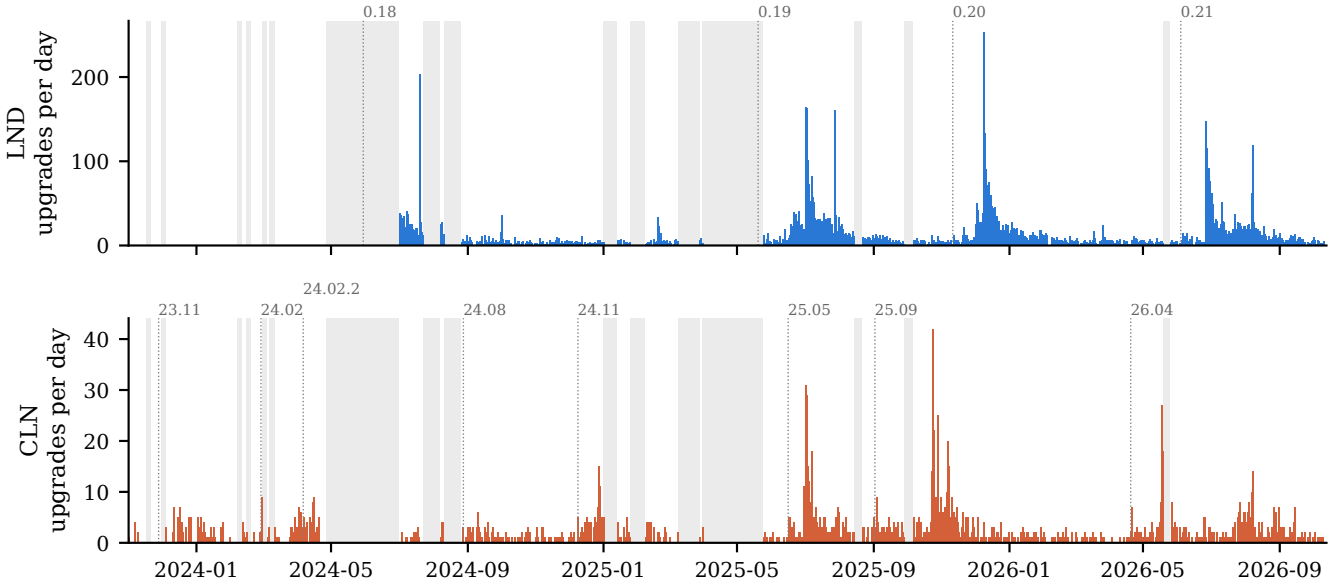


Fig. 5. Nodes first seen on a newer release line per day, where the upgrade day is known to within two days (managed fleet excluded). Grey: snapshot gaps; dotted lines: releases.

C. CLN operators responded

CLN's two memory-exhaustion vulnerabilities, disclosed on 19 July 2026, were fixed at the 25.09 and 26.04 line starts. Here the disclosure changed behaviour. Among live CLN nodes on lines before 26.04, the 30-day upgrade rate rose from 5.8% before to 24.0% after the disclosure (8.3% to 20.0% for lines before 25.09), and exposure kept falling after the maintainers' call on 26 August to upgrade or restart with `--offline`. Of 524 CLN nodes exposed to the gossipd vulnerability at disclosure, 243 had upgraded by the cutoff and 170 were no longer live, leaving 111; exposure fell from 66.6% of live CLN nodes to 24.2% (from 57.8% to 12.7% of capacity). Part of the fall is nodes leaving rather than upgrading, as in the shutdown we measured earlier [11].

VII. Discussion

a) Disclosure reaches the operators who were already upgrading: The LND advisories of 2025 and 2026 were published long after the fixes shipped, a common practice that lets operators upgrade before attackers learn what to look for. It works for the operators who follow releases: half of LND nodes move to a new line within three months, and the largest move first. The nodes still on an affected line at disclosure are the ones that did not, and the disclosure itself did not change their behaviour. Their upgrade rate in the month after Lightning Labs' batch was the same as in the

TABLE III

ADVISORIES WHOSE FIX BOUNDARY IS VISIBLE IN GOSSIP (LND AND CLN). EXPOSURE = SHARE OF LIVE NODES OF THAT IMPLEMENTATION WHOSE INFERRED RELEASE LINE IS AFFECTED, IN %; SHARE OF THEIR LIVE CAPACITY IN PARENTHESES; FOR PARTIAL ROWS (THE FIX LANDS INSIDE A LINE) THE BOUNDS ARE GIVEN IN BRACKETS AT DISCLOSURE. FIX→DISC. = DAYS FROM THE FIRST FIXED RELEASE TO PUBLIC DISCLOSURE. # = ADVISORIES GROUPED IN THE ROW.

Impl.	Advisory	Class	#	Fixed in	Fix→disc.	At fix	At disclosure	+60 d	Cutoff
CLN	Fake channel DoS (2023-08-23)	PARTIAL	1	23.02	175	–	–	6 (2)	1 (0)
CLN	Replacement cycling (CVE-2023-40234) (2023-10-16)	PARTIAL	1	23.08.1	34	–	38 [24–100] (17)	26 (10)	2 (0)
CLN	BOLT11 parsing bugs, batch 1 (2023-12-08)	PARTIAL	1	23.02	282	–	4 [4–95] (2)	2 (1)	1 (0)
CLN	BOLT11 parsing bugs, batch 2 (2023-12-08)	LINE	1	23.11	10	99 (100)	96 (96)	69 (73)	3 (0)
CLN	funding_created assertion DoS (2026-05-16)	LINE	1	25.09	256	98 (97)	29 (24)	24 (21)	12 (2)
CLN	connectd queue memory exhaustion (2026-07-19)	LINE	1	25.09	320	98 (97)	24 (20)	14 (3)	12 (2)
CLN	gossipd unknown-SCID memory exhaustion (2026-07-19)	LINE	1	26.04	90	100 (100)	67 (58)	29 (16)	24 (13)
CLN	Ping-flood OOM (2026-08-25)	LINE	1	25.09	357	98 (97)	19 (16)	–	12 (2)
LND	Fake channel DoS (2023-08-23)	PARTIAL	1	0.16.0	147	–	–	0 (0)	1 (0)
LND	Replacement cycling (CVE-2023-40233) (2023-10-16)	PARTIAL	1	0.17.0	13	–	0 [0–100] (0)	0 (0)	1 (0)
LND	Premature channel_update OOM (2023-12-21)	LINE	1	0.14.0	764	–	0 (0)	0 (0)	1 (0)
LND	channel_update signature after rate limit (2023-12-21)	PARTIAL	1	0.15.0	546	–	0 [0–0] (0)	0 (0)	1 (0)
LND	Onion bomb (CVE-2024-38359) (2024-06-18)	PARTIAL	1	0.17.0	259	–	–	–	1 (0)
LND	Irrevocable fees / update_fee breach fee-burn (2024-12-10)	PARTIAL	1	0.18.3	89	44 (35)	33 [33–100] (25)	24 (11)	4 (0)
LND	Excessive HTLC failback (2025-03-04)	LINE	1	0.18.0	278	–	20 (8)	–	4 (0)
LND	gossip_timestamp_filter goroutine DoS (2025-07-22)	PARTIAL	1	0.18.3	313	44 (35)	11 [11–60] (3)	9 (2)	4 (0)
LND	Replacement stalling, failback #2, infinite inbox (2025-12-04)	LINE	3	0.19.0	196	98 (96)	28 (12)	22 (10)	13 (6)
LND	Zero-timestamp gossip panic (2026-06-18)	PARTIAL	1	0.20.1	126	48 (73)	34 [34–97] (27)	25 (16)	20 (10)
LND	Lightning Labs batch: gossip, peer and queue DoS (2026-08-11)	LINE	6	0.19.0	446	98 (96)	15 (6)	13 (6)	13 (6)
LND	HTLC first-stage sweep budget failure (2026-08-11)	LINE	1	0.19.0	446	84 (92)	10 (5)	8 (6)	8 (6)
LND	Too few confirmations on cooperative close (2026-08-13)	PARTIAL	1	0.20.1	182	48 (73)	25 [25–51] (17)	21 (10)	20 (10)
LND	Gossip stall via malformed announcements (2026-09-21)	PARTIAL	1	0.20.1	221	27 (64)	8 [8–32] (5)	–	8 (4)
LND	Brontide write-allocation OOM (2026-09-21)	LINE	1	0.20.0	305	99 (99)	22 (11)	–	20 (10)

Before July 2024 our view did not include the few live nodes on LND releases older than 0.14 (§III); exposures to the 2023 LND advisories at disclosure are lower bounds. –: no usable snapshot within 7 days.

month before, and most of them were still exposed two months later. Once a release is a year old, the exposed tail consists of nodes that upgrade rarely or never, which an advisory alone does not reach. CLN's July disclosures did move operators: the rate at which nodes left vulnerable lines rose two- to fourfold in the following month, before the maintainers' call of 26 August, after which exposure fell further. We cannot separate the reasons. The fixed CLN lines were younger (one of the two fixes had shipped 90 days before disclosure, against 446 for LND), so more of the exposed operators were ones who usually keep up; CLN's operators are fewer and closer to its maintainers; and its disclosures came amid an active run of CLN security releases. Part of CLN's decline was nodes going offline rather than upgrading, which the maintainers asked for and which also removes the exposure.

b) Platforms move nodes; size and attention keep them current: Upgrades arrive in waves of a hundred or more nodes a few weeks after a release, consistent with node platforms shipping updates to all their users at once, and one wave followed a merchant platform's security release within a day. This matches what Frei et al. found for browsers: the update mechanism, more than the user, sets the patch level [6]. The tail is the opposite population: small nodes, nodes that skipped earlier releases, nodes no platform updates for them. For advisories that require a channel with the victim, these nodes are also the cheapest to reach. Their share of capacity at risk is small (about 6.4% for the August 2026 LND batch), but each holds its operator's funds.

c) Counting nodes: For half a year, one operator ran more than a fifth of the live LND nodes we observe, with almost no capacity. Any statistic that counts nodes, whether implementation shares, version shares or exposure, would have been dominated by that operator's choices. Fleets are cheap to create and hard to see; we found this one only because it upgraded in step. Node-level Lightning measurements should look for such fleets before counting, and report capacity alongside nodes [11].

VIII. LIMITATIONS

Release lines, not patches. Gossip cannot separate patch releases, so fixes inside a line, including all of CLN's 2026 patch releases, cannot be measured; for advisories with a fix inside a line we report bounds. *Inference.* An operator can switch features on or off so that an older line looks like a newer one or the reverse; the strict range covers every non-developer configuration, and validation on self-identifying CLN nodes bounds the error of the likely line, but we have no equivalent ground truth at scale for LND. A node that upgrades without re-announcing, or whose announcement we do not receive, keeps its old bits, which biases exposure up. *Vantage and liveness.* All data come from one node's view of the public graph; unannounced nodes, including most mobile wallets, are not covered, and liveness is inferred from peers' channel updates. *Fleets.* We identified one managed fleet by its alias and hubs. Smaller fleets without a shared alias would go undetected and be counted as independent nodes. *Exposure is not exploitability.* Several advisories require a channel with the victim, a particular configuration or inbound connections; we count nodes on affected releases, not nodes that can be attacked. Our advisory list is compiled from public sources and may miss silent fixes. *Association.* The hazard model describes which nodes upgrade sooner; it does not identify why.

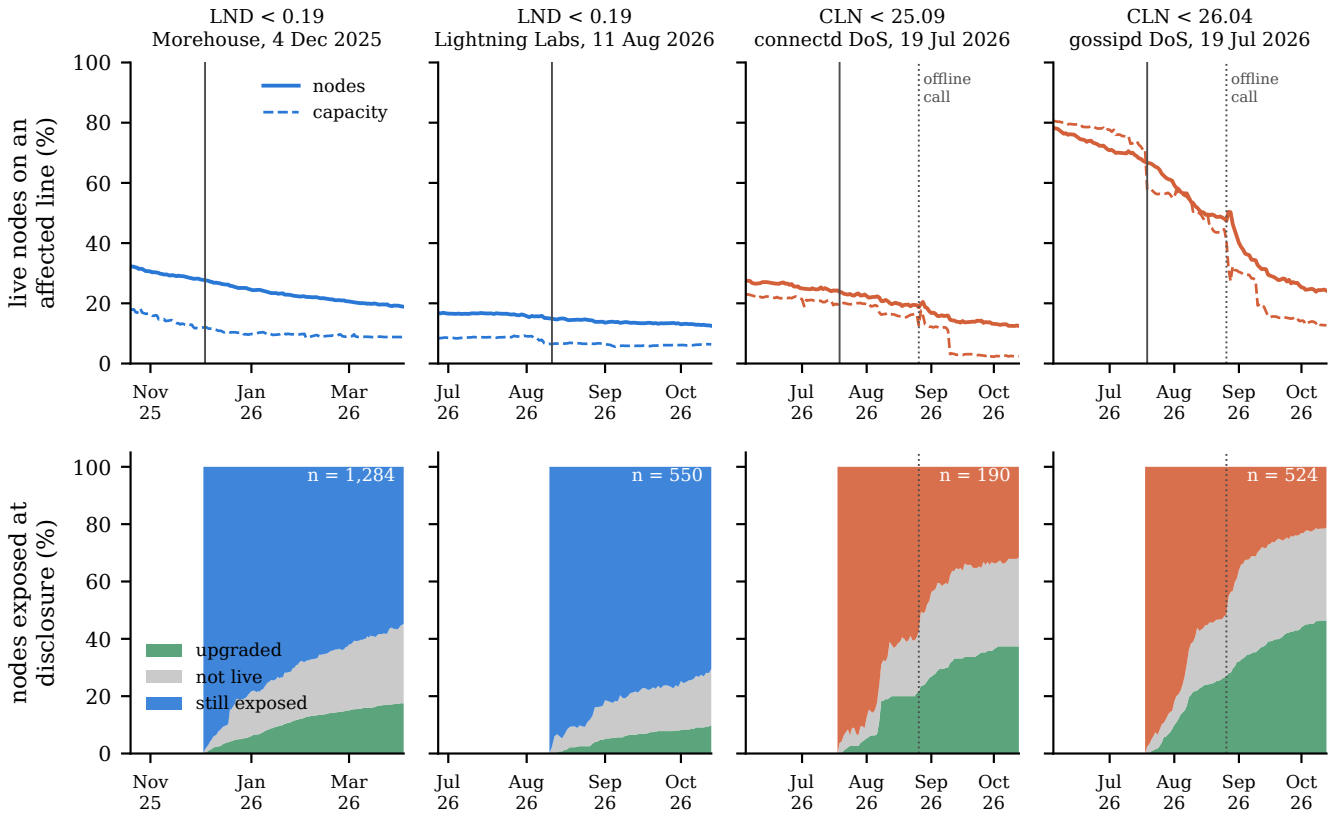


Fig. 6. Top: share of live nodes (solid) and of their live capacity (dashed) on a line affected by the advisory; solid vertical line: disclosure; dotted: CLN's call of 26 August 2026 to upgrade or restart offline. Bottom: the nodes exposed on the last snapshot before disclosure, by what became of them. Managed fleet excluded.

IX. Ethics

Release-line inference identifies nodes running vulnerable software, and in 2026 Lightning deployments were exploited (BTCPay Server [37]) and CLN maintainers reported attacks on unpatched nodes. We therefore publish aggregates only: no list, ranking, filter or label of individual nodes by release or exposure, in the paper or in the released data. All inputs are public gossip, every advisory we study had been published before our analysis began, and the paper reports no new vulnerability.

X. Conclusion

Lightning gossip carries no version string, but the feature bits each release announces identify its release line closely enough to measure how the network patches. Most operators upgrade within a few months, in waves pushed by their platforms; a tail of small and neglected nodes does not, and LND's disclosures, which typically come long after the fix, leave that tail where it was. CLN's 2026 disclosures, followed by the maintainers' appeal, moved theirs. Measured by capacity, the exposure is small; measured by nodes, it depends on whether one counts a single operator's fleet as hundreds of operators.

Data and Code Availability

Code, the aggregate datasets behind every figure and table, and a script that recomputes every number quoted in the text are available at https://github.com/bitcoindatalabs/ln-upgrade-dynamics. All stages truncate their inputs at the data cutoff of 13 October 2026. The underlying gossip data are public; our daily gossip snapshots are available from the author on request.

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